

Yixin (Amanda) Yin

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EDUCATION

University of Toronto

Sep 2022 - 2026

HBSc - Co-op Double Major in Computer Science and Statistics

Courses: Intro to Comp Sci I and II, Software Design, System Programming, ML and Data Mining, Numerical Analysis, Calculus I and II, Linear Algebra I and II, Probability I and II, Stochastic Process.

Awards: E-Fund Entrance Scholarship (10,000 CAD), DSI Research Scholar (7,200 CAD), Dean's List to date

TAships (Teacher Assistant): W24 CSCA67 (Discrete Mathematics), MATA37 (Calc 2).

SKILLS

Languages: C/C++, Rust, Java, Python, JavaScript/TypeScript, Go, HTML/CSS, R, SQL, GraphQL.

Tools/Frameworks: Git, Linux, MySQL, MongoDB, Power BI, AWS, React, Scikit-learn, PyTorch, MATLAB, Solr.

Skills: Client-Server Protocol and API Design, Data Modeling, Product Management, Microservices, Kubernetes.

INTERNSHIPS

Boosted.ai SWE Intern | *Fintech startup, Reliability Team, Toronto, ON*

Jan 2026 – April 2026

- Built an agent scoring pipeline with **guardrails** and **RAG**-based validation to improve reliability of AI workflows.
- Developed an abstract **gRPC** client library with centralized retry, timeout, and connection pooling configurations, improving service latency by up to **10×**.

Microsoft SWE Intern | *Outlook Group, Enterprise Adoption Team. Redmond, WA*

May-Aug 2025

- Integrated legacy **PST** (Personal Storage Table) data into the new Outlook by building a **native-to-web** bridge in **Rust** using MAPI, Win32GQL, and GraphQL to surface local task data in a modern React/TS environment.

StackAdapt SWE Intern | *Ad-Tech startup, Planning Forecast Team. Toronto, ON*

Jan-Apr 2025

- Led **full-stack** development of **Geo-Radius Forecasting** (TypeScript, Rails, GQL), including CSV ingestion, input validations, geolocation API integration, backend flow for S3 uploads, and live map UI rendering.
- Wrote a 15-page tech spec, scoped **30+** tickets, and coordinated weekly cross-team syncs for implementation.
- Hosted a **Tech Tune-Up** on SOLID principles for React best practices to elevate frontend standards.

Microsoft SWE and PM Intern | *Outlook Group, Enterprise Adoption Team, Redmond, WA*

May – Aug 2024

- Developed the Outlook app bar **MRU** and intelligent pinning spec, defining engineering requirements, user signals, telemetry metrics, and edge-case behavior for PM/stakeholder review.
- Implemented **MRU sorting and pinning logic** to capture, fetch, update, and query user acquisition signals; tuned prompt frequency and shipped production PRs with **500+ LOC**.

Scotiabank SWE and ML Research Intern | *Corporate AI/ML Analytics Team, Toronto, ON*

Jan – Apr 2024

- Built a full-stack analytics pipeline to generate unique **queryIDs**, linking user signals such as clicks and views into single search events and storing them in **MongoDB** for downstream analysis.
- Generated relevance **judgment lists** using the **Simplified Dynamic Bayesian Network** click model, engineered/logged Solr features, and implemented **pairwise SVM LTR** to automate to improve search relevance.

RESEARCH

Ultra-low-bit LLM Quantization | *Data Sciences Institute, University of Toronto*

May–Sep 2025

- Co-first-authored **ICAART 2026 Paper** on ultra-low-bit **LLM quantization** with sliced-Wasserstein distribution alignment; selected as **Best Paper** among 500+ accepted papers.

PAC-Bayes Safety Monitor | *University of Toronto, Department of Computer Science*

Jun 2026–Present

- Built a **Qwen3 + Aegis 2.0** safety-classification pipeline to study **LLM safety monitors** under reasoning vs. non-reasoning inference, prompt sensitivity, and safe/unsafe recall tradeoffs.
- Trained compressed adapters including **LoRA** and **LoRA-XS**, connecting adapter size and validation error to future **PAC-Bayes generalization bounds guarantees**.

Secure Agent Permission System | *Apart Research / supervised by Max von Hippel, Benchify (YC S24)*

Jun 2026–Present

- Researched a **formal agent-permission system** for LLM agents, modeling tool-use executions as finite traces and compiling high-level policies into **LTLf-style runtime monitors** that enforce tool-call, provenance, and benign-vs-adversarial behavior constraints.